

DAVIDE CORTINOVIS

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Google Scholar

EDUCATION

Center for Brain/Mind Sciences, University of Trento

PhD in Cognitive and Brain Sciences

Rovereto, Italy

2021 - April 2026 (expected)

My research is at the intersection of cognitive and computational neuroscience, focusing on two topics:

- Spatial organization of object dimensions in human high-level visual cortex: I collect and analyse fMRI data to study how functional selective areas are organized in the occipitotemporal cortex, especially focusing on the larger scale principles (animacy, real-world size, action) that drive their location and computations.
- Topographic models: I explore models that simulate the spatial and functional organization of the brain, aiming to test their convergence and divergence with visual cortex.
- My contribution to the field is the discovery of a large scale principle based on the action-related properties of objects that explains the arrangement of category-selective areas such as hands and tools. I also highlighted gaps in the ability of state-of-the-art neural networks to explain this action-based organization.

Justus Liebig University Giessen

Visiting PhD

Giessen, Germany

2024

I spent three months in Martin Hebart's lab as a visiting PhD student, where I analysed an fMRI dataset that I previously collected, compared it with a series of topographic artificial neural networks, and ran a series of encoding modelling analyses to test category selectivity in visual cortex.

Center for Brain/Mind Sciences, University of Trento

Master in Cognitive Neuroscience. Final grade: 110 with honors

Rovereto, Italy

2019 - 2021

University of Leeds

Erasmus exchange

Leeds, United Kingdom

2018-2019

University of Bergamo

Bachelor of Psychology. Final grade: 110 with honors

Bergamo, Italy

2016 - 2019

PUBLICATIONS

1. **Cortinovis, D.**, Truong, N., Op de Beeck, H., & Bracci, S. (2025). Investigating action topography in visual cortex and artificial neural networks. *Nature Communications*. [paper]
2. **Cortinovis, D.**, Peelen, M. V., & Bracci, S. (2025). Tool representations in human visual cortex. *Journal of Cognitive Neuroscience*, 37(3), 515-531. [paper]

CONFERENCE ABSTRACTS

1. **Cortinovis, D.**, Hebart, M., & Bracci, S. (2025). Comparing Object Selectivity in Visual Cortex and Topographic Deep Artificial Neural Networks. *Conference on Cognitive Computational Neuroscience*. [conference abstract]
2. **Cortinovis, D.**, Orlandi, G., van Campenhout, L., & Bracci, S. (2025). Object dimensions underlying food selectivity in visual cortex. *Journal of Vision*, 25(9), 1910-1910.
3. Bracci, S., **Cortinovis, D.**, & Garnuto, E. (2025). The Effect of Scene Clutter on Visual Representations. *Journal of Vision*, 25(9), 2061-2061.

4. **Cortinovis, D.**, Op de Beeck, H., & Bracci, S. (2023). The role of action-related properties in shaping the object space in the biological and artificial brain. *Journal of Vision*, 23(9), 4868-4868.
5. **Cortinovis, D.**, Op de Beeck, H., & Bracci, S. (2021). The organization of body-parts representations in Deep Convolutional Neural Networks. *Perception*, 50(1_SUPPL), 123-123.

POSTER AND TALK PRESENTATIONS

1. Object dimensions underlying food selectivity in visual cortex. **Talk** at Vision Sciences Society, St. Pete Beach, Florida, USA (May 2025). [\[link\]](#)
2. Category-selectivity, again: Hand-selective clusters in visual cortex support action-related behavioural goals. **Talk** at Organization for Human Brain Mapping (OHBM), Seoul, South Korea (June 2024).
3. The action and animacy feature spaces reveal a functional dissociation between lateral and ventral occipitotemporal cortex. Poster presentation at Concepts, Action, Objects (CAOs) Workshop, Rovereto, Italy (May 2024)
4. Category-selectivity, again: Hand-selective clusters in visual cortex support action-related behavioural goals. **Talk** at TEAP, Regensburg, Germany (March 2024).
5. Category-selectivity, again: Hand-selective clusters in visual cortex support action-related behavioural goals. Poster presentation at EWCN, Bressanone, Italy (January 2024).
6. Unveiling the Dimensions of Action Object Space: Insights from fMRI and DCNNs. Poster Presentation at Neuro-AI Talks, Osnabrueck, Germany (September 2023)
7. Unveiling the Dimensions of Action Object Space: Insights from fMRI and DCNNs. Poster presentation at Seeing and Acting Workshop (SAW), Coimbra, Portugal (September 2023)
8. The role of action-related properties in shaping the object space in the biological and artificial brain. Poster presentation at Concepts, Action, Objects (CAOs) Workshop, Rovereto, Italy (May 2023)
9. The role of action-related properties in shaping the object space in the biological and artificial brain. Poster presentation at ECVP, Nijmegen, Netherlands (August 2022).

ACADEMIA SERVICES

- Reviewer for *Imaging Neuroscience*.
- Reviewer for *Cortex*.

TEACHING AND MENTORSHIP

- Teaching assistant for the course Computational Modeling of Perception for the master program in Cognitive Science at the University of Trento (2024/2025 and 2025/2026). My part of the course focused on spatial organization in visual, auditory, and language cortex and how to model that with (topographic) artificial neural networks.
- Teaching assistant for the course Research Design and Hands-on Statistics for the master program in Cognitive Science at the University of Trento (2022/2023).
- Supervision of internship and thesis of four master students.

AWARD

- Fully-funded scholarship for the Master program at the University of Trento, 2019-2021.

ADDITIONAL INFORMATION

Languages	Italian (native), English (C1), Spanish (B1), French (B1)
Technical skills	MATLAB, Python, fMRI (data collection and data analysis)